
Statistics 212: Lecture 16 (April 2, 2025)

Construction of Ito Integral

Instructor: Mark Sellke

Scribe: Aniket Jain

1 Constructing the Itô Integral

Goal: Define

$$\int_0^\infty H(t) dB(t),$$

where $H(t) = H(t, \omega)$ is a progressively measurable process. Concretely, we require

$$H: [0, T] \times \Omega \rightarrow \mathbb{R}$$

to be measurable with respect to $\mathcal{B}(\mathbb{R}) \otimes F_T$ (or more precisely, progressively measurable) and

$$\mathbb{E} \left[\int_0^\infty H(t, \omega)^2 dt \right] < \infty.$$

1.1 Definition on simple processes.

A *simple process* $H(t, \omega)$ has the form

$$H(t, \omega) = \sum_{i=1}^k A_i(\omega) \mathbf{1}_{(t_i, t_{i+1}]}(t),$$

for a partition $0 = t_0 < t_1 < \dots < t_k = T$, where each $A_i(\omega)$ is F_{t_i} -measurable. Then we define

$$\int_0^\infty H(t, \omega) dB_t = \sum_{i=1}^k A_i(\omega) [B_{t_{i+1}} - B_{t_i}].$$

Key property: Itô $\mathcal{L}^2$ isometry.

On simple processes, we have

$$\mathbb{E} \left[\left(\int_0^T H(t, \omega) dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H(t, \omega)^2 dt \right].$$

1.2 Approximation by Simple Processes

Fact 16.1 Let H be a progressively measurable process such that

$$\mathbb{E}\left[\int_0^\infty H(t, \omega)^2 dt\right] < \infty.$$

Then there exists a sequence of simple processes $(H_n)_{n \geq 1}$ such that

$$\mathbb{E}\left[\int_0^\infty (H_n(t) - H(t))^2 dt\right] \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Proof (outline):

(a) *Truncating the time domain.*

Choose T large enough so that

$$\mathbb{E}\left[\int_T^\infty H(t, \omega)^2 dt\right] \leq \varepsilon.$$

Then it suffices to approximate $H(t, \omega) \mathbf{1}_{[0, T]}(t)$, which we denote by $H^{(1)}(t, \omega)$.

(b) *Clipping the range.*

Define the *clip* function

$$\text{Clip}_N(x) = \min(N, \max(-N, x)).$$

Choose N large enough so that

$$\int_0^T [H^{(1)}(t, \omega) - \text{Clip}_N(H^{(1)}(t, \omega))]^2 dt \leq \varepsilon.$$

Set $H^{(2)}(t, \omega) = \text{Clip}_N(H^{(1)}(t, \omega))$. Existence of such T and N follows from the dominated convergence theorem.

(c) *Step-function approximation in time.*

Partition $[0, T]$ using $t_k = \frac{k}{2^n} T$ for $k = 0, 1, 2, \dots, 2^n T$. Then define

$$H_n^{(3)}(t, \omega) = 2^n \int_{t_k}^{t_{k+1}} H^{(2)}(s, \omega) ds \quad \text{for } t \in [t_k, t_{k+1}).$$

Each $H_n^{(3)}$ is again progressively measurable (adapted and right-continuous) (as long as $H^{(2)}$ is progressively measurable).

By the Lebesgue differentiation theorem, or the following martingale argument, one may show $H_n^{(3)} \rightarrow H^{(2)}$ in L^2 , as $n \rightarrow \infty$.

Define the filtration

$$\mathcal{G}_n = \sigma\left(\{[0, 2^{-n}], [2^{-n}, 2 \cdot 2^{-n}], \dots\}, \mathcal{F}\right).$$

Then

$$\mathbb{E}[H(t, \omega) \mid \mathcal{G}_n] = 2^n \int_{t_k}^{t_{k+1}} H(t, \omega) dt \quad \text{for } t \in [t_k, t_{k+1}).$$

Define the *Doob martingale* $(M_n)_{n \geq 1}$ by

$$M_n = \mathbb{E}[H \mid \mathcal{G}_n] \quad \text{for } H \in \mathcal{L}^2.$$

Then

$$M_n \xrightarrow{\text{a.s., } \mathcal{L}^2} \mathbb{E}[H \mid (\mathcal{G}_1, \mathcal{G}_2, \dots)],$$

where the latter is the conditional expectation w.r.t. $\mathcal{B}(\mathbb{R}) \otimes F$

1.3 The Itô integral

(i) *Setup.* Let H be progressively measurable with $\mathbb{E}\left[\int_0^\infty H(t, \omega)^2 dt\right] < \infty$. Take a sequence of simple processes $(H_n)_{n \geq 1}$ such that $\mathbb{E}\left[\int_0^\infty (H - H_n)^2 dt\right] \rightarrow 0$.

(ii) *Definition.* Define

$$\int_0^\infty H(t) dB_t = \lim_{n \rightarrow \infty} \int_0^\infty H_n(t) dB_t \quad (\text{limit in } \mathcal{L}^2).$$

(a) *Existence.* Show that $\{\int_0^\infty H_n dB_t\}_n$ is Cauchy in L^2 .

(b) *Uniqueness.* The limit cannot depend on the particular approximating sequence $H_n \rightarrow H$.

Hence:

(i) $\int_0^\infty H_n dB_t \rightarrow \int_0^\infty H dB_t$ in L^2 ,

(ii) The limit is the same for any other sequence $\tilde{H}_n \rightarrow H$,

(iii) $\mathbb{E}[(\int_0^\infty H dB_t)^2] = \mathbb{E}[\int_0^\infty H(t)^2 dt]$.

(i) Apply L^2 isometry.

We have

$$\mathbb{E}\left[\left(\int_0^\infty H_n dB_t - \int_0^\infty H_m dB_t\right)^2\right] = \mathbb{E}\left[\int_0^\infty (H_n - H_m)^2 dt\right] \xrightarrow{n, m \rightarrow \infty} 0.$$

Thus $\{\int_0^\infty H_n dB_t\}$ is Cauchy in $\mathcal{L}^2$ and hence convergent in $\mathcal{L}^2$.

(ii) Uniqueness via interleaving.

If $\{H_n\} \rightarrow H$ in L^2 and $\{\tilde{H}_n\} \rightarrow H$ in L^2 , then the interleaved sequence $H_1, \tilde{H}_1, H_2, \tilde{H}_2, \dots$ also converges to H . Their Itô integrals must converge to the same limit, ensuring uniqueness of the definition $\int_0^\infty H dB_t$.

(iii) $\mathcal{L}^2$ Isometry:

By construction

$$\int_0^\infty H dB_t := \lim_{n \rightarrow \infty} \int_0^\infty H_n dB_t \quad (\text{in } L^2).$$

Hence the L^2 norms converge, establishing the isometry property.

2 Time progressive Itô integral

We set

$$\int_0^t H(s) dB_s = \int_0^\infty H(s) \mathbf{1}_{[0, t]}(s) dB_s,$$

allowing us to discuss the structure of the stochastic process $M(t) = \int_0^t H(s) dB_s$ as t varies.

Key tool: L^2 maximal inequality.

Theorem 16.1 Let H be progressively measurable with $\mathbb{E}\left[\int_0^\infty H(s)^2 ds\right] < \infty$

Then there exists an *almost surely continuous modification* of $\{M_t = \int_0^t H(s) dB_s : t \geq 0\}$ which is also a martingale.

Modification: Two stochastic processes X_t and X'_t are said to be *modifications* of each other if for all t , $\mathbb{P}(X_t \neq X'_t) = 0$

Proof:

Case: Simple processes

If $H(s, \omega)$ is *simple*, say

$$H(s, \omega) = \sum_{i=1}^k A_i(\omega) \mathbf{1}_{(t_i, t_{i+1}]}(s),$$

then

$$\int_0^t H(s) dB_s = \sum_{i=1}^k A_i \left(B_{t_{i+1} \wedge t} - B_{t_i \wedge t} \right),$$

which is

- a.s. continuous (as BM is a.s. continuous)
- a martingale
- progressively measurable

Case: General H

Approximate $H_n \xrightarrow{L^2} H$, define

$$M_n(t) = \int_0^t H_n(s) dB_s,$$

By Doob's L^2 Maximal Inequality

$$\mathbb{E} \left[\left(\sup_{0 \leq t < \infty} |M_n(t) - M_{n'}(t)| \right)^2 \right] \leq 4 \mathbb{E} \left[(M_n(\infty) - M_{n'}(\infty))^2 \right] = 4 \mathbb{E} \left[\int_0^\infty (H_n(s) - H_{n'}(s))^2 ds \right] \rightarrow 0$$

Fix $1 < \alpha < 2$

By Chebyshev,

$$\mathbb{P} \left(\sup_{t \geq 0} |M_n(t) - M_{n'}(t)| > \alpha^{-k} \right) \leq 4 \alpha^{2k} \mathbb{E} \left[\int_0^\infty (H_n(s) - H_{n'}(s))^2 ds \right] = \left(\frac{\alpha^2}{4} \right)^k \quad (\text{for } n, n' > n_k \text{ large}).$$

It follows that $\{M_{n_k} : k \geq 0\}$ is a.s. Cauchy in $C([0, \infty), d_{sup})$ for suitable subsequence $n_1 < n_2 < \dots$

Because we may choose $\{n_k\}$ so that

$$\mathbb{P} \left[\sup_{t \geq 0} |M_{n_{k+1}}(t) - M_{n_k}(t)| \geq \alpha^{-k} \right] \leq \left(\frac{\alpha^2}{4} \right)^k$$

which is summable.

Hence by Borel-Cantelli Lemma, all these events hold past random $\bar{k} = \bar{k}(\omega)$. If $k, k' > \bar{k}$,

$$\sup_t |M_{n_k}(t) - M_{n_{k'}}(t)| \leq \sum_{j \geq \min(k, k')} \alpha^{-j} \rightarrow 0$$

as $k, k' \rightarrow \infty$

Thus, M_{n_k} converge a.s. to an a.s. continuous limit $M'(t)$, which is a modification of $M(t) = \int_0^t H(s) dB_s$.